

Lecture Notes on Numerical Methods

IIITDM Kancheepuram, Chennai, India

07 January 2025

Exhaustive Search Method for Root Finding

Numerical Methods

Finding roots of nonlinear equations

$$f(x) = 0$$

Given a continuous function $f(x)$, find a value $x = \alpha$ such that

$$f(\alpha) = 0$$

- ▶ Exact analytical solutions may not exist.
- ▶ Hence numerical root-finding methods are required.

Goal

Approximate the root within a prescribed accuracy ε .

Idea of Exhaustive Search Method

- ▶ Divide the interval $[a, b]$ into small subintervals.
- ▶ Evaluate the function at equally spaced points.
- ▶ Look for a change of sign of $f(x)$.

$$f(x_i) f(x_{i+1}) < 0 \Rightarrow \text{root lies in } (x_i, x_{i+1})$$

Key Principle

Based on the Intermediate Value Theorem.

Algorithm: Exhaustive Search Method

Step 1: Choose initial interval $[a, b]$ such that $f(a)f(b) < 0$.

Step 2: Select step size h .

Step 3: Compute points:

$$x_i = a + ih, \quad i = 0, 1, 2, \dots$$

Step 4: Evaluate $f(x_i)$.

Step 5: If

$$f(x_i)f(x_{i+1}) < 0,$$

then a root lies in (x_i, x_{i+1}) .

Step 6: Stop when interval width $\leq \varepsilon$.

Accuracy and Computational Cost

- ▶ Interval width after search: h
- ▶ Approximate root error:

$$|\alpha - x| \leq \frac{h}{2}$$

Function Evaluations

$$N = \frac{b - a}{h}$$

To achieve accuracy ε : $h \leq 2\varepsilon$

$$\Rightarrow N = O\left(\frac{1}{\varepsilon}\right)$$

Advantages and Limitations

Advantages

- ▶ Very simple to understand.
- ▶ Guaranteed to find a root if sign change exists.
- ▶ No derivative required.

Limitations

- ▶ Computationally expensive.
- ▶ Slow convergence.
- ▶ May miss multiple or closely spaced roots.

Bisection Method

Idea:

- ▶ Start with interval $[a, b]$ such that:

$$f(a)f(b) < 0$$

- ▶ Repeatedly halve the interval.
- ▶ Select the subinterval containing the root.

Interval Reduction

Interval length is reduced by half at each iteration.

Bisection (Interval Halving) Method

Algorithm for Bisection Method:

Input: $a, b, M, \delta, \epsilon$

$u \leftarrow f(a)$

$v \leftarrow f(b)$

$e \leftarrow b - a$

Output: a, b, u, v

if $\text{sign}(u) = \text{sign}(v)$ **then**

end if

for $k = 1$ **to** M **do**

$e \leftarrow e/2$

$c \leftarrow a + e$

$w \leftarrow f(c)$

Output: k, c, w, e

if $|e| < \delta$ **or** $|w| < \epsilon$ **then**

end if

if $\text{sign}(w) \neq \text{sign}(u)$ **then**

$b \leftarrow c$

$v \leftarrow w$

else

$a \leftarrow c$

$u \leftarrow w$

end if

end for

Bisection Method

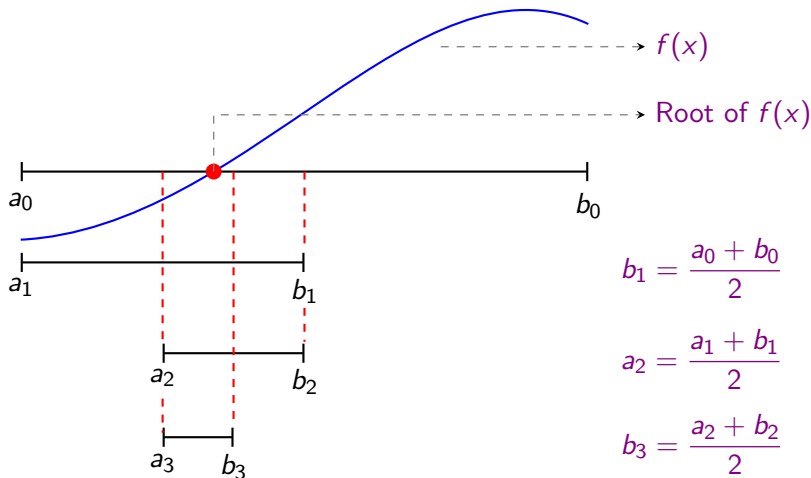


Figure 1: Graphical representation of the bisection method.

Computational Cost: Bisection Method

After k iterations:

$$\text{Interval length} = \frac{b - a}{2^k}$$

To achieve accuracy ε :

$$\frac{b - a}{2^k} \leq \varepsilon$$

$$k \geq \log_2 \left(\frac{b - a}{\varepsilon} \right)$$

Complexity

$$N = O \left(\log_2 \frac{1}{\varepsilon} \right)$$

Computational Efficiency Comparison

Methods Compared:

Exhaustive Search

- ▶ Evaluates objective function at uniformly spaced points.
- ▶ Number of function evaluations & For accuracy ε

$$N = \frac{b-a}{\Delta}; \text{ and } N = O\left(\frac{1}{\varepsilon}\right)$$

Interval Halving Method

- ▶ Reduces search interval by half at each iteration.
- ▶ After k iterations & for accuracy ε

$$\text{Interval length} = \frac{b-a}{2^k}; \text{ and } k = O\left(\log_2 \frac{b-a}{\varepsilon}\right)$$

Conclusion

Interval halving is exponentially more computationally efficient than exhaustive search.

Computational Efficiency Comparison

Criterion	Exhaustive Search	Bisection Method
Search strategy	Uniform scanning	Interval halving
Function evaluations	$O(1/\varepsilon)$	$O(\log(1/\varepsilon))$
Convergence rate	Linear	Logarithmic
Speed	Very slow	Fast
Guarantee of convergence	Yes	Yes
Derivative required	No	No

Key Observation

Efficiency Insight

Exhaustive search reduces error linearly, whereas bisection reduces error exponentially.

$$\text{Exhaustive: } \varepsilon \propto \frac{1}{N} \quad \text{Bisection: } \varepsilon \propto 2^{-N}$$

Conclusion

Bisection method is computationally far more efficient than exhaustive search.

Error Analysis

Bisection Method generates successive interval

$$[a_0, b_0]$$

$$[a_1, b_1]$$

$$[a_2, b_2]$$

$$\vdots$$

$$[a_n, b_n]$$

such that

$$a_0 \leq a_1 \leq a_2 \leq \cdots \leq b_0$$

$$b_0 \geq b_1 \geq b_2 \geq \cdots \geq a_0$$

and

$$b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n) \quad (n \geq 0) \quad (1)$$

Error Analysis Contd.

- ▶ Sequence $\{a_n\}$ is non-decreasing and bounded above by b_0 , it converges.
- ▶ Sequence $\{b_n\}$ is non-increasing and bounded below by a_0 , it converges.

If we apply Eq(1) repeatedly, we find that

$$(b_n - a_n) = 2^{-n}(b_0 - a_0)$$

Thus

$$\lim_{n \rightarrow \infty} b_n - \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} 2^{-n}(b_0 - a_0) = 0$$

If we put

$$r = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n$$

Error Analysis Contd.

Then by taking a limit in the inequality

$$0 \geq f(a_n)f(b_n),$$

we obtain $0 \geq [f(r)]^2$.

Hence

$$f(r) = 0$$

Suppose that, the process stopped at $[a_n, b_n]$ then **Estimated Root** is

$$c_n = \frac{a_n + b_n}{2}$$

The error is then bounded as

$$|r - c_n| \leq \frac{1}{2}(b_n - a_n) = 2^{-(n+1)}(b_0 - a_0).$$

Therom on Bisection Method

If $[a_0, b_0]$, $[a_1, b_1]$, $[a_2, b_2]$, $\dots$, $[a_n, b_n]$, $\dots$ denote the intervals in the bisection method, then the limits $\lim_{n \rightarrow \infty} a_n$ and $\lim_{n \rightarrow \infty} b_n$ exist, are equal, and represent a zero of f .

If $r = \lim_{n \rightarrow \infty} c_n$ and $c_n = \frac{1}{2}(a_n + b_n)$, then

$$|r - c_n| \leq 2^{-(n+1)}(b_0 - a_0) \quad (2)$$



Example:

Suppose that the bisection method is started with the interval $[50, 63]$. How many steps should be taken to compute a root with relative accuracy of one part in 10^{-12} ?

Solution:

Relative accuracy

$$\left| \frac{r - c_n}{r} \right| \leq 10^{-12}$$

we know

$$r \geq 50 \implies \frac{1}{r} \leq \frac{1}{50}$$

For finding n , it suffices to secure the inequality,

$$\frac{|r - c_n|}{50} \leq 10^{-12}$$

Replace $|r - c_n|$ by Eq(2),

$$2^{-(n+1)} \times \frac{13}{50} \leq 10^{-12}$$

Solving this for n , we conclude that $n \geq 37$



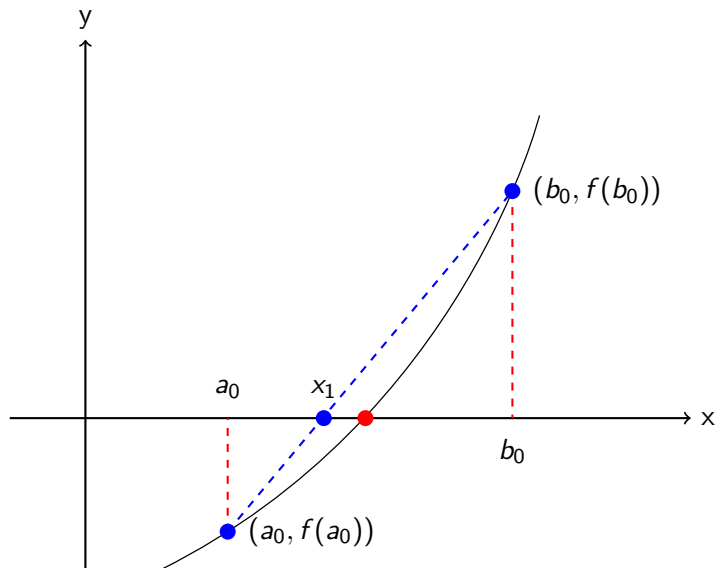
Exercise

1. Determine the formula

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log a_0}{\log 2} - 1$$

involving a_0 , b_0 and ϵ for the number of steps that should be taken in the bisection algorithm to ensure that root is determined with *relative accuracy* $\leq \epsilon$. Assume $a_0 > 0$.

The Method of False Position(Regula - Falsi Method)



The Method of False Position(Regula - Falsi Method)

The method of false position uses the x - intercept of the line which passes through the points $(a_0, f(a_0))$ and $(b_0, f(b_0))$.

The equation of the line which passes through $(a_0, f(a_0))$ and $(b_0, f(b_0))$ is given by

$$y = f(a_0) + \frac{f(b_0) - f(a_0)}{b_0 - a_0}(x - a_0).$$

To find x co-ordinate, put $y = 0$.

$$f(a_0) + \frac{f(b_0) - f(a_0)}{b_0 - a_0}(x_1 - a_0) = 0$$

where x_1 is first approximation.

The Method of False Position(Regula - Falsi Method)

$$x_1 = a_0 - f(a_0) \frac{b_0 - a_0}{f(b_0) - f(a_0)}$$

Which can also be written as

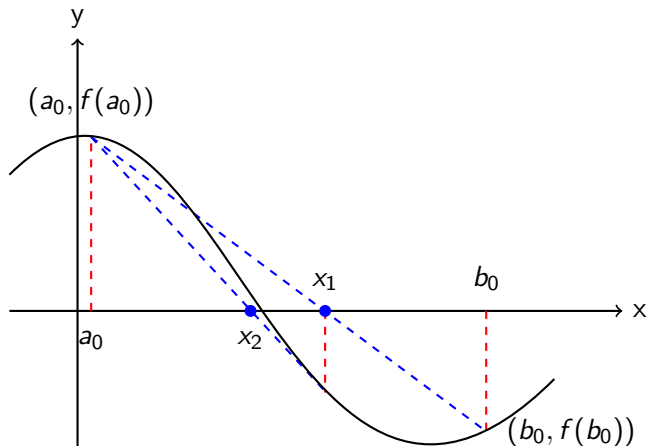
$$x_1 = \frac{a_0 f(b_0) - b_0 f(a_0)}{f(b_0) - f(a_0)}$$

There are two cases.

Case I: If x_1 is the solution of non linear equation $f(x) = 0$ i.e. $f(x_1) = 0$ then stop.

Case II: Either $f(a_0)f(x_1) < 0$ or $f(b_0)f(x_1) < 0$

The Method of False Position(Regula - Falsi Method)



The Method of False Position(Regula - Falsi Method)

The **Case II** turns up then,

$$[a_1, b_1] = \begin{cases} [a_0, x_1] & \text{if } f(a_0)f(x_1) < 0 \\ [x_1, b_0] & \text{if } f(b_0)f(x_1) < 0 \end{cases}$$

General Term,

$$x_{n+1} = a_n - f(a_n) \frac{b_n - a_n}{f(b_n) - f(a_n)}$$

Or,

$$x_{n+1} = \frac{a_n f(b_n) - b_n f(a_n)}{f(b_n) - f(a_n)}$$

Prove that:

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} b_n = r(\text{root})$$

Example

Consider non - linear equation $e^x - 2 = 0$. Verify Regula - falsi for $a_0 = 0, b_0 = 1$.

Solution:

Since $f(a_0)f(b_0) = f(0)f(1) = -1 \cdot (e - 2) < 0$.

Iteration 1:

$$\begin{aligned}x_1 &= \frac{a_0 f(b_0) - b_0 f(a_0)}{f(b_0) - f(a_0)} \\&= \frac{1}{(e - 2) - (1 - 2)} \\&= \frac{1}{e - 1} \\&\approx 0.582\end{aligned}$$

To determine whether the zero is contained on $(a_0, x_1) = (0, 0.582)$ or on $(x_1, b_0) = (0.582, 1)$, we calculate

$$f(x_1) = -0.2104 \leq 0$$

Solution contd.

since $f(a_0)f(x_1) \geq 0$ and $f(b_0)f(x_1) \leq 0$. So, zero is between x_1 and b_0 . For the next iteration, we therefore take $(a_1, b_1) = (x_1, b_0) = (0.582, 1)$.

Iteration 2:

$$x_2 \approx 0.67669$$

$$f(x_2) \approx -0.032645$$

Calculate up to **iteration 3**.

Exercise: The function $f(x) = \sin x$ has a zero on the interval $(3, 4)$, namely $x = \pi$. Perform the four iterations of the methods of the false position to approximate zero.



Convergence Analysis

Here, we have **two natural questions**.

- ▶ Does the sequence of approximations generated by the method of false position, p_n , converge to a root p ?
- ▶ What is the order of convergence?

To **answer these questions**,

We need to examine the error sequences $\{e_n\}$, where $e_n = p_n - p$. Since,

$$\{p_n\} \rightarrow p \iff \{e_n\} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

Now,

$$p_n - p = b_n - p - f(b_n) \cdot \frac{b_n - a_n}{f(b_n) - f(a_n)}$$

Convergence Analysis contd.

Now we approximate the function values $f(a_n)$ and $f(b_n)$ by second-degree Taylor polynomials.

$$f(a_n) \approx f'(p)(a_n - p) + \frac{f''(p)}{2}(a_n - p)^2,$$

$$f(b_n) \approx f'(p)(b_n - p) + \frac{f''(p)}{2}(b_n - p)^2,$$

Where the fact that $f(p) = 0$ as p is root.

$$\begin{aligned} f(b_n) - f(a_n) &\approx f'(p)(b_n - a_n) + \frac{f''(p)}{2}[(b_n - p)^2 - (a_n - p)^2] \\ &= (b_n - a_n) \left[f'(p) + \frac{f''(p)}{2}(b_n + a_n - 2p) \right] \end{aligned}$$

Convergence Analysis contd.

Now recall,

$$p_n - p = (b_n - p) - f(b_n) \frac{b_n - a_n}{f(b_n) - f(a_n)}$$

Substituting the value $f(b_n)$ and $(f(b_n) - f(a_n))$

$$\begin{aligned} p_n - p &\approx (b_n - p) - \frac{(b_n - a_n) \left[f'(p)(b_n - p) + \frac{f''(p)}{2}(b_n - p)^2 \right]}{(b_n - a_n) \left[f'(p) + \frac{f''(p)}{2}(b_n + a_n - 2p) \right]} \\ &= (b_n - p) \left[1 - \frac{f'(p) + \frac{f''(p)}{2}(b_n - p)}{f'(p) + \frac{f''(p)}{2}(b_n + a_n - 2p)} \right] \end{aligned}$$

Convergence Analysis contd.

$$p_n - p \approx (b_n - p)(a_n - p) \frac{f''(p)}{2f'(p) + f''(p)(b_n + a_n - 2p)} \quad (3)$$

Eq.(3) is called error evolution equation. Now Eq.(3) becomes

$$e^n \approx \lambda e_{n-1}$$

where

$$\lambda = \frac{lf''(p)}{2f'(p) + lf''(p)} \quad (4)$$

and

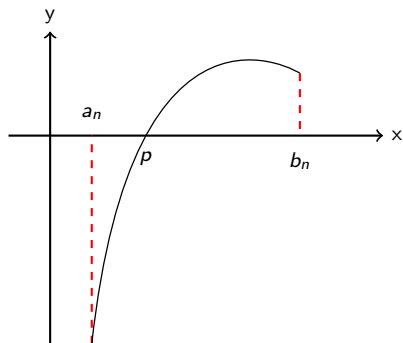
$$l = \begin{cases} a_n - p, & a_n \text{ remains fixed} \\ b_n - p, & b_n \text{ remains fixed} \end{cases}$$

Convergence Analysis contd.

- ▶ If $|\lambda| < 1$, then it follows that the error sequence (e^n) generated by the false position method converges.
- ▶ **Convergence is linear** with **asymptotic error constant** λ .
- ▶ The key role of the proof is: how to bring $p_n - p$ in RHS.
- ▶ Let's try to understand, how a_n or b_n be p_{n-1} in it's previous iteration.

Convergence Analysis contd.

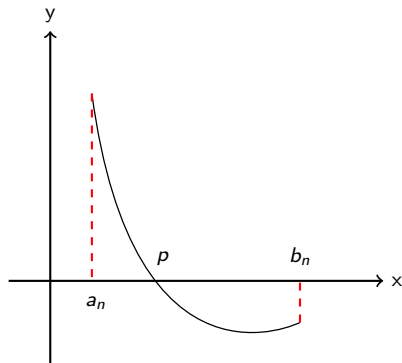
Figure 1



$$f'(p) > 0, \quad f''(p) < 0$$

$$a_n \text{ is fixed, } b_n = p_{n-1}$$

Figure 2

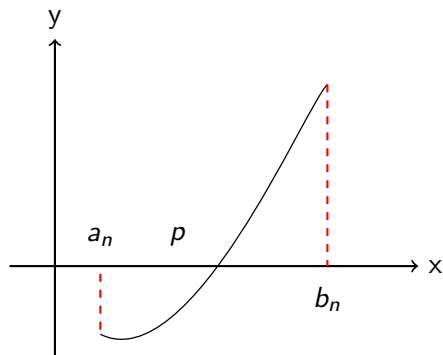


$$f'(p) < 0, \quad f''(p) > 0$$

$$a_n \text{ is fixed, } b_n = p_{n-1}$$

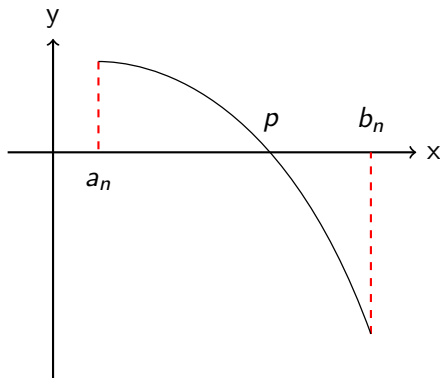
Convergence Analysis contd.

Figure 3



$$f'(p) > 0, \quad f''(p) > 0$$
$$b_n \text{ is fixed, } a_n = p_{n-1}$$

Figure 4



$$f'(p) < 0, \quad f''(p) < 0$$
$$b_n \text{ is fixed, } a_n = p_{n-1}$$

Convergence Analysis contd.

From the **Figure 1**, a_n is fixed, so $l = a_n - p < 0$ and since $f''(p) < 0$, we have $(a_n - p)f''(p) > 0$.

$$\begin{aligned} 2f'(p) + (b_n + a_n - 2p)f''(p) &> 2f'(p) + (a_n - p)f''(p) \\ \because (b_n - p) &> 0 \end{aligned}$$

So,

$$\lambda = \frac{(a_n - p)f''(p)}{2f'(p) + (a_n - p)f''(p)} < 1$$

Hence

$$|\lambda| < 1$$

Exercise: Show that $|\lambda| < 1$ for remaining three figures.



Test example 1

$$f(x) = \tan(\pi x) - x - 6 \text{ on } (0.4, 0.48)$$

n	a_n	b_n	Approximation
1	0.4000000000	0.4800000000	0.4208674108
2	0.4208674108	0.4800000000	0.4332027501
3	0.4332027501	0.4800000000	0.4404957388
4	0.4404957388	0.4800000000	0.4448079249
5	0.4448079249	0.4800000000	0.4473577484
6	0.4473577484	0.4800000000	0.4488655162
7	0.4488655162	0.4800000000	0.4497571072
8	0.4497571072	0.4800000000	0.4502843380

Table 1: b_n is fixed and $a_n \approx p_n$

Test example 2.

$$f(x) = x^3 + 2x^2 - 3x - 1 \text{ on } (-3, -2)$$

n	a_n	b_n	Approximation
1	—3.0000000000	—2.0000000000	—2.8333333333
2	—3.0000006000	—2.8333333333	—2.9079283887
3	—3.0000000000	—2.9079283887	—2.9120026293
4	—3.0000000000	—2.9120026293	—2.9122172667
5	—3.0000000000	—2.9122172667	—2.9122285522
6	—3.0000000000	—2.9122285522	—2.9122291456
7	—3.0000000000	—2.9122291456	—2.9122291768
8	—3.0000000000	—2.9122291768	—2.9122291784

Table 2: a_n is fixed and $b_n \approx p_n$

Stopping condition

Terminate Iteration when $|e_n| < \epsilon$

$$\begin{aligned}e_n &= p_n - p \\&= p_n - p_{n-1} + p_{n-1} - p \\&= p_n - p_{n-1} + e_{n-1} \\&= p_n - p_{n-1} + \frac{1}{\lambda} e_n \quad \because (e_n = \lambda e_{n-1})\end{aligned}$$

$$\left(1 - \frac{1}{\lambda}\right) \cdot e_n = p_n - p_{n-1}$$

$$|e_n| \approx \left| \frac{\lambda}{1 - \lambda} \right| [p_n - p_{n-1}] < \epsilon$$

So, we terminate the condition when

$$\left| \frac{\lambda}{1 - \lambda} \right| [p_n - p_{n-1}] < \epsilon \quad (5)$$

Stopping condition contd.

Now, How we estimate the value of λ ?

Consider the ratio

$$\begin{aligned}\frac{p_n - p_{n-1}}{p_{n-1} - p_{n-2}} &= \frac{(p_n - p) - (p_{n-1} - p)}{(p_{n-1} - p) - (p_{n-2} - p)} \\&= \frac{e_n - e_{n-1}}{e_{n-1} - e_{n-2}} \\&\approx \frac{(\lambda - 1)e_{n-1}}{(1 - \frac{1}{\lambda})e_{n-1}} \quad (\because e_n = \lambda e_{n-1}, e_{n-2} = \frac{1}{\lambda} e_{n-1}) \\&= \lambda\end{aligned}$$

Exercise: By using the method of false position to approximate the all real roots of $f(x) = x^3 - x^2 - 10x + 7$. Uses an absolute tolerance of 10^{-6} as stopping condition.

Error estimate

$$f(x) = x^3 + 2x^3 - 3x - 1 \text{ on } (1, 2)$$

Iteration (n)	Absolute Error	Error Estimate
	(e_n)	$\frac{\lambda}{1-\lambda}[p_n - p_{n-1}]$
1	0.0986912435	—
2	0.0469476054	—
3	0.0218503335	9.0236382347
4	0.0100635702	0.0104374516
5	0.0046123322	0.0046903760
6	0.0021091553	0.0021254290
7	0.0009634891	0.0009668808
8	0.0004399257	0.0004406324
9	0.0002008251	0.0002009723
10	0.0000916671	0.0000916978

Table 3: Absolute Error and Error Estimate for Each Iteration

Fixed Point Iteration

Basic definition and theroms :

Definition(Fixed point:)

A real number p is called a **fixed point** of g if $g(p) = p$.

Example:

$x = 0$ is fixed point for $\sin x$ because $\sin 0 = 0$

Example (Fixed point and root)

Suppose $\alpha \neq 0$. Show that p is a root of $f(x)$ iff p is a fixed point of

$$g(x) := x - \alpha f(x)$$

Fixed Point Theorem

Theorem:

1. If $g \in C[a, b]$ and $a \leq g(x) \leq b$ for all $x \in [a, b]$, then g has at least one fixed point in $[a, b]$.
2. If, in addition, g' exists in $[a, b]$, and $\exists k < 1$ such that $|g'(x)| \leq k < 1$ for all x , then g has a unique fixed point in $[a, b]$.

Proof: 1

- ▶ If $g(a) = a$ or $g(b) = b$, then done.
- ▶ Otherwise, $g(a) > a$ and $g(b) < b$.
- ▶ Define $f(x) = x - g(x)$, then $f(a) = a - g(a) < 0$, and $f(b) = b - g(b) > 0$.
- ▶ By IVT and since f is continuous, $\exists p \in (a, b)$ such that $f(p) = 0$, i.e., $p - g(p) = 0$.

Example

Proof: 2

- ▶ If $\exists p, q \in [a, b]$ are two distinct fixed points of g , then $\exists \xi \in (p, q)$ such that

$$1 = \frac{p - q}{p - q} = \left| \frac{g(p) - g(q)}{p - q} \right| = |g'(\xi)| \leq k < 1$$

by MVT. Contradiction.

Example:

Let $f(x) = \cos(x)$ on $[0, 1]$.

- ▶ $f(x)$ is continuous on $[0, 1]$.
- ▶ Since $\cos(x) \in [0, 1]$ for $x \in [0, 1]$, $f : [0, 1] \rightarrow [0, 1]$.
- ▶ By the fixed-point theorem, $f(x) = \cos(x)$ has at least one fixed point in $[0, 1]$.

Numerically solving $\cos(x) = x$ gives a fixed point near $x \approx 0.739$.

Example (Application of Fixed Point Theorem)

$g(x) = \frac{x^2-1}{3}$ has a unique fixed point in $[-1, 1]$.

- ▶ First we need to show $g(x) \in [-1, 1]$, $\forall x \in [-1, 1]$. Find the max and min values of g as $-\frac{1}{3}$ and 0 (Hint: find critical points of g first).
- ▶ So $g(x) \in [-\frac{1}{3}, 0] \subset [-1, 1]$.
- ▶ Also $|g'(x)| = |\frac{2x}{3}| \leq \frac{2}{3} < 1$, $\forall x \in [-1, 1]$, so g has unique fixed point in $[-1, 1]$ by FPT.
- ▶ The fixed point is $p = \frac{3 - \sqrt{13}}{2} = -0.302776$.

Example (Fixed Point Theorem – Failed Case 1)

- ▶ $g(x) = \frac{x^2-1}{3}$ has a unique fixed point in $[3, 4]$. But we can't use FPT to show this.
- ▶ Note that there is a unique fixed point in $[3, 4]$ ($p = \frac{3+\sqrt{13}}{2}$), but $g(4) = 5 \notin [3, 4]$, and $g'(4) = \frac{8}{3} > 1$ so we cannot apply FPT here.
- ▶ From this example, we know FPT provides a *sufficient* but not *necessary* condition.

Example (Fixed Point Theorem – Failed Case 2)

- ▶ We can use FPT to show that $g(x) = 3^{-x}$ must have FP on $[0, 1]$, but we can't use FPT to show if it's unique (even though the FP on $[0, 1]$ is unique in this example).
- ▶ **Solution.** $g'(x) = (3^{-x})' = -3^{-x} \ln 3 < 0$, therefore $g(x)$ is strictly decreasing on $[0, 1]$. Also $g(0) = 3^0 = 1$ and $g(1) = 3^{-1}$, so $g(x) \in [0, 1]$, $\forall x \in [0, 1]$. So a FP exists by FPT.
- ▶ However, $g'(0) = -\ln 3 \approx -1.098$, so we do not have $|g'(x)| < 1$ over $[0, 1]$. Hence FPT does not apply.
- ▶ Nevertheless, the FP must be unique since g strictly decreases and intercepts with $y = x$ line only once.

Fixed Point Iteration

Theorem: Let $g : [a, b] \rightarrow [a, b]$ be a function such that

- (i) $g(x)$ is continuous on $[a, b]$
- (ii) $g(x)$ is differentiable on (a, b)
- (iii) $\exists k \in (0, 1)$ such that $|g'(x)| \leq k < 1, \forall x \in (a, b)$

Then,

1. Sequence $\{p_n\} \rightarrow p$, where $p_n = g(p_{n-1})$ for any $p_0 \in [a, b]$;
2. $|p_n - p_{n-1}| \leq k^n \max(p_0 - a, b - p_0)$; and,
3. $|p_n - p| \leq \frac{k^n}{1-k} |p_1 - p_0|$.

Remark 1: Interval $[a, b]$ must be closed and bounded.

Remark 2: All three given conditions guarantee that there exist one and only one fixed point p in $[a, b]$.

Proof

(1)

$$\begin{aligned}|p_n - p| &= |g(p_{n-1}) - g(p)| && (\because p = g(p), p_n = g(p_{n-1})) \\&= |g'(\zeta)| |p_{n-1} - p| && (\text{Mean Value Theorem}) \\&\leq k |p_{n-1} - p| && \text{where } |g'(\zeta)| \leq k < 1 \\&\leq k^2 |p_{n-2} - p| \\&\vdots \\&\leq k^n |p_0 - p|\end{aligned}$$

since $k < 1$

$$\begin{aligned}\lim_{n \rightarrow \infty} |p_n - p| &\leq \lim_{n \rightarrow \infty} k^n |p_0 - p| \\&= (p_0 - p) \lim_{n \rightarrow \infty} k^n \\&= 0\end{aligned}$$

$$\therefore p_n \rightarrow p \text{ as } n \rightarrow \infty$$

Proof contd.

(2)

$$\begin{aligned} |p_n - p| &\leq k^n |p_0 - p| \\ \implies |p_n - p| &\leq k^n \max(p_0 - a, p_0 - b) \end{aligned}$$

(3)

$$\begin{aligned} |p_{n+1} - p_n| &= |g(p_n) - g(p_{n-1})| \\ &= |g'(\zeta)| |p_n - p_{n-1}| \\ &\leq k |p_n - p_{n-1}| \\ &\vdots \\ &\leq k^n |p_1 - p_0| \end{aligned}$$

Now, let $m > n$. Then,

Proof contd.

$$\begin{aligned} |p_m - p_n| &= |p_m - p_{m-1} + p_{m-1} - p_{m-2} + \cdots + p_{n+1} - p_n| \\ &\leq |p_m - p_{m-1}| + |p_{m-1} - p_{m-2}| + \cdots + |p_{n+1} - p_n| \\ &\leq k^{m-1}|p_1 - p_0| + k^{m-2}|p_1 - p_0| + \cdots + k^n|p_1 - p_0| \\ &= k^n|p_1 - p_0|(1 + k + k^2 + \cdots + k^{m-n-1}) \end{aligned}$$

since, $p_m \rightarrow p$ as $m \rightarrow \infty$ then,

$$\begin{aligned} |p - p_n| &\leq k^n|p_1 - p_0| \sum_{i=0}^{\infty} k^i \\ |p_n - p| &\leq \frac{k^n}{1 - k}|p_1 - p_0|, \end{aligned} \tag{6}$$

Importance of k : When k is small, the error decreases quickly, so order of convergence becomes higher but as k approaches 1, the order of convergence becomes much slower. □

Example

Consider the function $f(x) = x^3 + x^2 - 3x - 3$. Now we have to find the zero of $f(x)$ in $(1, 2)$. ($\because f(1)f(2) = -4 \cdot 3 < 0$)
ie

$$f(x) = 0$$

$$x^3 + x^2 - 3x - 3 = 0$$

$$\frac{x^3 + x^2 - 3}{3} = x$$

$$g(x) = x$$

So, here zero of $f(x)$ and fixed point of $g(x)$ are same.

Remark: Every rootfinding problem can be transformed into multiple fixed point problems. Some of these fixed point problems converge rapidly, some will converge slowly and some will not converge at all.

Example contd.

$f(x) = x^3 + x^2 - 3x - 3 = 0$ can be written as following fixed point problems.

$$g_1(x) = \frac{1}{3}(x^3 + x^2 - 3)$$

$$g_2(x) = -1 + \frac{3x + 3}{x^2}$$

$$g_3(x) = \sqrt[3]{3 + 3x - x^2}$$

$$g_4(x) = \sqrt{\frac{3 + 3x - x^2}{x}}$$

$$g_5(x) = x - \frac{x^3 + x^2 - 3x - 3}{3x^2 + 2x - 3}.$$

Now, we are going to check which of the above problems converge or diverges, generated by sequence $p_n = g(p_{n-1})$ and $p_0 = 1$.

Example contd.

n	$g_1(x)$	$g_2(x)$	$g_3(x)$
0	+1.0	+1.0	+1.0
1	-0.3333333333	5.0000000000	1.709975947
2	-0.9753086420	-0.2800000000	1.733134316
3	-0.9921709716	26.5510204100	1.731994802
4	-0.9974310264	-0.8827544117	1.732053695
5	-0.9991480696	-0.5486245114	1.732050659
6	-0.9997165068	3.4989256110	1.732050815
7	-0.9999055558	0.1024544340	1.732050807
8	-0.9999685246	314.0796731	-
9	-0.9999895088	-0.9904178717	-
10	-0.9999965030	-0.9706946913	-

Table 4: $g_1(x)$ converges to $-1 \notin (1, 2)$, $g_2(x)$ diverges, and $g_3(x)$ converges rapidly.

Example contd.

n	$g_4(x)$	$g_5(x)$
0	+1.0	+1.0
1	2.236067978	-3.000000000
2	1.451059202	-2.200000000
3	1.901682432	1.830150754
4	1.635808067	1.737795453
5	1.788336635	1.732072292
6	1.699764653	1.732050808
7	1.750767137	1.732050808
8	1.721269132	—
9	1.738283891	—
10	1.738454854	—

Table 5: $g_4(x)$ converge slowly while $g_5(x)$ converges rapidly

Question: How to know which g_i converges to its fixed point ?

Order Of convergence for FPI

Theorem: Let $g : [a, b] \rightarrow [a, b]$ be a function such that

- (i) $g(x)$ is continuous on $[a, b]$.
- (ii) $g'(x)$ is continuous on (a, b) .
- (iii) $|g'(x)| \leq k < 1, \forall x \in (a, b)$

Claim: If $g'(p) \neq 0$, then for any $p_0 \in [a, b]$ the sequence generated by $p_n = g(p_{n-1})$ converges linearly to the fixed point p .

Proof: Only claim :

$$g'(p) \neq 0 \implies \text{convergence is linear.}$$

OR

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|} = \lambda \in (0, 1) \text{ for some } \lambda.$$

Proof contd.

Consider,

$$\begin{aligned}|p_{n+1} - p| &= |g(p_n) - g(p)| \\ &= |g'(c_n)| |p_n - p|,\end{aligned}$$

Also given g is continuous on (a, b) , then

$$\lim_{n \rightarrow \infty} |g'(c_n)| = |g'(\lim_{n \rightarrow \infty} c_n)| = |g'(p)|.$$

Hence,

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|} = |g'(c_n)| = \lambda \in (0, 1) \text{ for some } \lambda.$$

$$(\because |g'(c_n)| \in (0, 1))$$

Hence the **convergence is linear**. (because power of $|p_n - p|$ is 1.)



Higher order of convergence

Theorem: Let $g : [a, b] \rightarrow [a, b]$ be a function such that

- (i) $g(x)$ is continuous on $[a, b]$
- (ii) for $\alpha > 1$, $g', g'', \dots, g^{\alpha-1}, g^\alpha$ are continuous in (a, b) .
- (iii) Let $p \in (a, b)$ be a fixed point of g and $\exists$ a α such that $g'(p) = g''(p) = \dots = g^{\alpha-1}(p) = 0$ but $g^\alpha(p) \neq 0$

Claim:

- ▶ There exists a $\delta > 0$ such that for any $p_0 \in [p - \delta, p + \delta]$, the sequence $p_n = g(p_{n-1})$ converges to fixed point p .
- ▶ Sequence $p_n = g(p_{n-1})$ converges to fixed point p of order α and asymptotic error constant is

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^\alpha} = \frac{|g^\alpha(p)|}{\alpha!}$$

Proof

First item of the claim:

Since g' is continuous and $g'(p) = 0$. ($\because \alpha > 1$)

Now we use the definition of continuity.

For a given $0 < \epsilon < 1$, there exists a $\delta > 0$ such that

$$|g'(x) - g'(p)| < \epsilon \quad \text{for all } x \in (p - \delta, p + \delta).$$

Thus, we obtain

$$|g'(x)| < \epsilon \quad \text{for all } x \in (p - \delta, p + \delta).$$

So, $g'(x) \leq k < 1$ for $x \in (p - \delta, p + \delta)$. (take $k < \epsilon$)

Proof contd.

Set $I = [p - \delta, p + \delta]$

Now we show that $g : I \rightarrow I$.

For any $x \in I$, we apply Mean Value Theorem.

$$\begin{aligned} |g(x) - p| &= |g(x) - g(p)|, \quad \text{for } x \in [p - \delta, p + \delta], \quad p = g(p) \\ &= |g'(\zeta)| |x - p|, \quad \text{by MVT} \\ &\leq k |x - p| \\ &< |x - p| \quad (\because 0 < k < 1) \\ &\leq \delta \end{aligned}$$

So

$$\begin{aligned} -\delta &\leq g(x) - p \leq \delta \\ p - \delta &\leq g(x) \leq p + \delta \\ g(x) &\in I = [p - \delta, p + \delta], \quad \text{for } x \in I \end{aligned}$$

Then by fixed point iteration theorem, sequence $p_n = g(p_{n-1})$ converges to p for any $p_0 \in [p - \delta, p + \delta]$

Proof contd.

Second item of the claim:

Now we expand the $g(x)$ by Taylor series expansion about the point p in interval I .

$$g(x) = g(p) + g'(p)(x - p) + \cdots + \frac{g^{(\alpha-1)}(p)}{(\alpha-1)!}(x - p)^{\alpha-1} \\ + \frac{g^{(\alpha)}(\zeta)}{\alpha!}(x - p)^{\alpha} \text{ where } \zeta \in (x, p).$$

Also again $g^{(k)}(p) = 0$ for $0 \leq k \leq \alpha - 1$ and letting $x = p_n$, then Taylor series expansion simplifies to

$$p_{n+1} - p = \frac{g^{(\alpha)}(\zeta)}{\alpha!}(p_n - p)^{\alpha} \tag{7}$$

$$(\because g(x) = g(p_n) = p_{n+1} \text{ and } g(p) = p)$$

Proof contd.

Here $\zeta \in (p_n, p)$, so taking limit $n \rightarrow \infty$,

Then

$$p_n \rightarrow p, \text{ forcing } \zeta \rightarrow p$$

Hence, from Eq(6)

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^\alpha} = \frac{|g^\alpha(p)|}{\alpha!}.$$

$$\text{Order of convergence} = \alpha$$

$$\text{Asymptotic error constant} = \frac{|g^\alpha(p)|}{\alpha!}$$



Nearest Point Approximation of Root

Given that $f(x)$ is continuous function in $[a, b]$ and $f(a)f(b) < 0$.

Let r be the zero of $f(x)$ and x is the approximation of r and h is the error then $r = x + h$ and

$$f(r) = f(x + h) = 0$$

If $f''(x)$ is exists and continuous, then by Taylor's theorem,

$$0 = f(r) = f(x + h) = f(x) + hf'(x) + O(h^2)$$

That is $0 \approx f(x) + hf'(x)$ and $h \approx -\frac{f(x)}{f'(x)}$.

If x is an approximation to the root r then $x + h$ is an better approximation to r . This leads to an iteration scheme.

$$x_{n+1} = x_n + h \quad \text{or} \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Newton-Raphson Method

The x_n is the approximation at iteration number n , then the iteration scheme is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

This is named as Newton-Raphson Method.

Using the definition of $f'(x)$ given below

$$f'(x) = \lim_{u \rightarrow x} \frac{f(x) - f(u)}{x - u}$$

The derivative at x_n is given by

$$f'(x_n) = \lim_{h \rightarrow 0} \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

Secant Method

So,

$$f'(x_n) \approx \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}$$

Substituting the value of $f'(x_n)$ in Eq.(8), we get:

$$x_{n+1} = x_n - f(x_n) \cdot \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

Simplifying further:

$$x_{n+1} = \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Secant Method

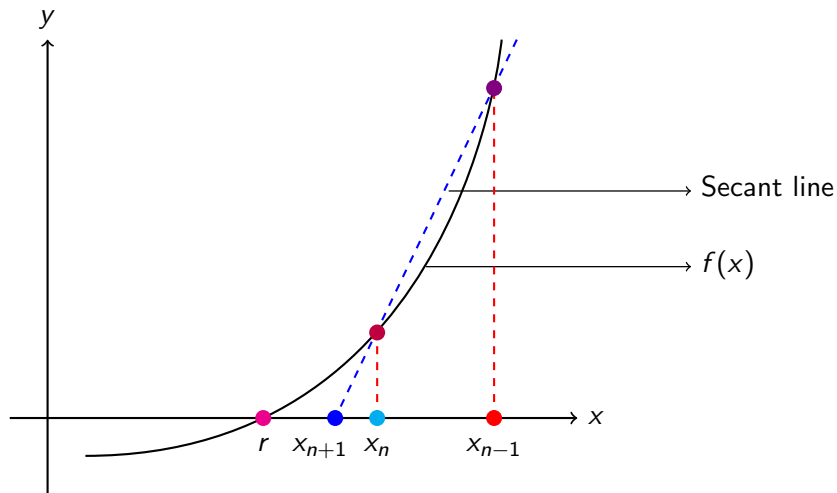


Figure 2: Graphical representation of the secant method.

Example:

Find zero of $f(x)$ by using of secant method.

$$f(x) = x^3 - \sinh x + 4x^2 + 6x + 9$$

Solution : Since $f(7)f(8) < 0$, so there is a zero between 7 and 8.
Take $x_0 = 8$ and $x_1 = 7$.

n	x_n	$f(x_n)$
0	8.00000	-0.665×10^3
1	7.00000	0.417×10^2
2	7.05895	0.208×10^2
3	7.11764	-0.183×10^1
4	7.11289	0.710×10^{-1}
5	7.11306	0.244×10^{-3}
6	7.11306	0.191×10^{-4}

Table 6: Convergence table for x_n and $f(n_n)$

Secant Algorithm

Input: $a, b, M, \delta, \epsilon$

$fa \leftarrow f(a)$

$fb \leftarrow f(b)$

Output: $0, a, fa$

Output: $1, b, fb$

for $k = 2$ to M **do**

if $|f(a)| > |f(b)|$ **then**

$a \leftrightarrow b; fa \leftrightarrow fb$

end if

$s \leftarrow (b - a)/(fb - fa)$

$b \leftarrow a$

$fb \leftarrow fa$

$a \leftarrow a - fa * s$

$fa \leftarrow f(a)$

Output: k, a, fa

if $|fa| < \epsilon$ or $|b - a| < \delta$ **then**

end if

end for

Error Analysis

Recall,

$$|e_{n+1}| = A|e_n|^\alpha$$

where A is a positive constant, $\alpha > 0$ is order of convergence and

$$e_n = x_n - r$$

Aim : We have to calculate the value of α .

$$e_{n+1} = x_{n+1} - r$$

$$= \frac{x_{n-1}f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})} - r \quad \text{from Eq(10)}$$

$$= \frac{(e_{n-1} + r)f(x_n) - (e_n + r)f(x_{n-1})}{f(x_n) - f(x_{n-1})} - r$$

$$= \frac{e_{n-1}f(x_n) + \cancel{rf(x_n)} - e_nf(x_{n-1}) - \cancel{rf(x_{n-1})} - \cancel{rf(x_n)} + \cancel{rf(x_{n-1})}}{f(x_n) - f(x_{n-1})}$$

Error Analysis Contd.

From the definition, we have:

$$\begin{aligned}e_{n+1} &= \frac{e_{n-1}f(x_n) - e_nf(x_{n-1})}{f(x_n) - f(x_{n-1})} \\&= \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \left[\frac{f(x_n)/e_n - f(x_{n-1})/e_{n-1}}{x_n - x_{n-1}} \right] e_n e_{n-1}. \quad (8)\end{aligned}$$

Here, we multiply and divide by $x_n - x_{n-1}$ and factor out $e_n e_{n-1}$.

By Taylor's Theorem, we have:

$$\begin{aligned}f(x_n) &= f(r + e_n) = f(r) + e_nf'(r) + \frac{1}{2}e_n^2f''(r) + O(e_n^3), \\ \frac{f(x_n)}{e_n} &= f'(r) + \frac{1}{2}e_nf''(r) + O(e_n^2).\end{aligned} \quad (9)$$

$$(\because f(r) = 0)$$

Error Analysis Contd.

Changing the index n to $n - 1$, we have:

$$\frac{f(x_{n-1})}{e_{n-1}} = f'(r) + \frac{1}{2}e_{n-1}f''(r) + O(e_{n-1}^2). \quad (10)$$

We subtract Eq. (10) from Eq. (9), then:

$$\frac{f(x_n)}{e_n} - \frac{f(x_{n-1})}{e_{n-1}} = \frac{1}{2}(e_n - e_{n-1})f''(r) + O(e_{n-1}^2).$$

Since $e_n - e_{n-1} = (e_n + r) - (e_{n-1} + r) = x_n - x_{n-1}$

$$\frac{f(x_n)/e_n - f(x_{n-1})/e_{n-1}}{x_n - x_{n-1}} \approx \frac{1}{2}f''(r). \quad (11)$$

Error analysis contd.

Again from Taylor's theorem

$$f(x_n) = f(r + e_n) = f(r) + e_n f'(r) + O(e_n^2),$$

$$\text{and } f(x_{n-1}) = f(r + e_{n-1}) = f(r) + e_{n-1} f'(r) + O(e_{n-1}^2),$$

By subtraction of these equations, we get

$$\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} \approx \frac{1}{f'(r)}. \quad \text{replace } n \quad (12)$$

$$(\because e_n - e_{n-1} = x_n - x_{n-1})$$

Error analysis contd.

Putting the value Eq.(11) and Eq. (12) in Eq. (8). Hence, we have

$$e_{n+1} \approx \frac{1}{2} \frac{f''(r)}{f'(r)} e_n e_{n-1} = C e_n e_{n-1} \quad (13)$$

Now,

$$\begin{aligned} |e_n| &= A |e_{n-1}|^\alpha \\ |e_{n-1}| &= \left(\frac{|e_n|}{A} \right)^{\frac{1}{\alpha}} \end{aligned}$$

Putting the value of e_{n+1} and e_{n-1} in Eq.(16)

$$A |e_n|^\alpha = |C| |e_n| \left(\frac{|e_n|}{A} \right)^{\frac{1}{\alpha}}$$

Error Analysis Contd.

$$A^{1+\frac{1}{\alpha}}|C|^{-1} = |e_n|^{1-\alpha+\frac{1}{\alpha}}$$

Since the left hand side is a nonzero constant when $e_n \rightarrow 0$. So $1 - \alpha + \frac{1}{\alpha} = 0$ because if $1 - \alpha + \frac{1}{\alpha} \neq 0$ then right hand side becomes zero as $e_n \rightarrow 0$ which implies either A or C is zero which is contradiction.

So,

$$1 - \alpha + \frac{1}{\alpha} = 0$$
$$\alpha = \frac{1 + \sqrt{5}}{2} \approx 1.62$$

Hence, the secant method's rate of convergence is **superlinear** (that is, better than liner).

Asymptotic error constant $A = ?$

$$A^{1+\frac{1}{\alpha}}|C|^{-1} = |e_n|^{1-\alpha+\frac{1}{\alpha}}$$

$$A^{1+\frac{1}{\alpha}}|C|^{-1} = 1 \quad (\because 1 - \alpha + \frac{1}{\alpha} = 0)$$

$$A = |C|^{\frac{1}{1+\frac{1}{\alpha}}}$$

$$A = |C|^{\frac{1}{\alpha}} = |C|^{\alpha-1} = |C|^{0.62} \quad (\because 1 + \frac{1}{\alpha} = \alpha)$$

so,

$$A = \left| \frac{f''(r)}{2f'(r)} \right|^{0.62}$$



The Difference between the secant method and false-position method

False-position method :

$$x_{n+1} = \frac{a_n f(b_n) - b_n f(a_n)}{f(b_n) - f(a_n)}$$

Where a_n and b_n is the end point of interval $[a_n, b_n]$ that we got in previous iteration.

Secant method:

$$x_{n+1} = \frac{x_{n-1} f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Where x_n and x_{n-1} are the last two iteration values. If we replace $a_n \rightarrow x_{n-1}$ and $b_n \rightarrow x_n$ then we get, both are **same formula**.

But what are the differences ?

Critical difference

- ▶ **Bracketing the root:** The false-position method always keeps the root between a_n and b_n , while the secant method does not guarantee.
- ▶ **Replacement Rule:** In the false position method, the new estimate x_{n+1} replaces a_n if $f(x_{n+1})f(a_n) > 0$ or x_{n+1} replaces b_n if $f(x_{n+1})f(b_n) > 0$. In the secant method the new value x_{n+1} replaces the value x_n and x_n replaces x_{i-1} .
- ▶ **Convergence:** The false-position method always converges because the root is kept within the bracket(Interval). The secant method can diverge if both values lie on the same side of root.
- ▶ **Stability:** The false-position method is more stable but may converge slower. The secant method is faster but can be less reliable.

Counter Example:

Example: Use the false-position and secant methods to estimate the root of $f(x) = \ln x$. Start the computation with values of $a_n = x_{i-1} = 0.5$ and $b_n = x_i = 5.0$.

Solution:

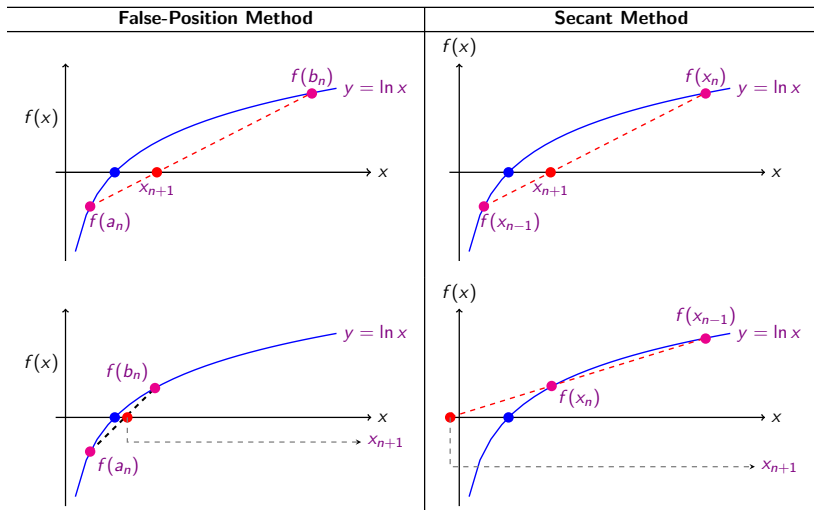
Iteration table for false position method:

Iteration	a_n	b_n	x_{n+1}
1	0.5	5.0	1.8546
2	0.5	1.8546	1.2163
3	0.5	1.2163	1.0585

Iteration table for secant method :

Iteration	x_{n-1}	x_n	x_{n+1}
1	0.5	5.0	1.8546
2	5.0	1.8546	-0.104 38

Comparison by graph of $f(x) = \ln x$



Comparison of the true percent relative error ϵ_t for the methods to determine the root of $f(x) = e^x - x$.

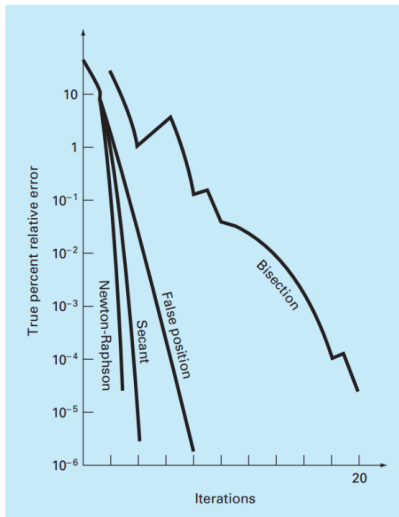


Figure 3: graph of $f(x) = e^x - x$.